

Consistency (ACID Property)

1. Overview

Consistency means a transaction must take the database from **one valid state to another valid state**.

None

Valid State → Transaction → Valid State

If data was consistent before execution, it must remain consistent after execution.

2. Simple Meaning

All defined rules, constraints, and integrity conditions must remain true after the transaction.

Examples of rules:

- No negative balance
- Unique email IDs
- Foreign keys valid
- Required fields present
- Total money preserved
- Stock cannot go below zero

3. Banking Example

Transfer ₹50 from Account A to Account B.

Before transaction:

None

A = 500

B = 300

Total = 800

After transaction:

None

A = 450

B = 350

Total = 800

Total remains same.

Database is consistent.

4. Inconsistent State Example

Suppose ₹50 deducted from A but not added to B:

None

A = 450

B = 300

Total = 750

₹50 disappeared.

This violates business rules and creates inconsistent state.

5. What Consistency Protects

Consistency ensures:

- Constraints remain valid
- Relationships remain correct
- No invalid partial business state
- Data integrity preserved

6. Common Consistency Rules in Databases

A. Primary Key Uniqueness

No duplicate IDs.

B. Foreign Key Validity

Order must reference existing user.

C. Check Constraints

None

```
balance >= 0
```

```
age >= 18
```

```
quantity >= 0
```

D. Business Rules

None

total debit = total credit

seat count cannot exceed capacity

7. SQL Example

SQL

```
CREATE TABLE accounts (  
    balance INT CHECK (balance >= 0)  
);
```

Transaction causing negative balance should fail.

8. Consistency vs Atomicity

Property	Meaning
Atomicity	All or nothing
Consistency	Valid state before and after

Atomicity ensures full execution.
Consistency ensures rules remain satisfied.

9. Consistency vs CAP Consistency

Important interview distinction:

ACID Consistency

Rules/integrity constraints preserved.

CAP Consistency

All nodes see same latest data (linearizability).

These are **different meanings**.

Mention this in interviews.

10. Real-World Ecommerce Example

Order placement rules:

None

`stock >= 0`

`payment captured once`

`order has valid user`

`invoice total correct`

Transaction must preserve these.

11. How Databases Enforce Consistency

- Constraints
- Triggers
- Transactions
- Referential integrity
- Schema validation
- Application business logic

Examples:

- PostgreSQL
- MySQL

12. Distributed Systems Consistency Example

If inventory updates replicate late:

None

Node A says stock=1

Node B says stock=5

That is distributed consistency issue (CAP context), different from ACID rule consistency.

13. Interview Answer Example

Q: What is consistency in ACID?

Consistency means every committed transaction preserves database rules and constraints, moving data from one valid state to another valid state.

14. Design Tip

Put critical rules in multiple layers:

None

Application validation

+ Database constraints

+ Transaction logic

15. Short Revision Notes

None

Consistency =

Valid state $\rightarrow$ Valid state

Examples:

- total money unchanged
- no negative balance
- valid foreign keys
- unique IDs

16. One-Line Summary

Consistency ensures every successful transaction preserves all database rules, constraints, and integrity requirements, keeping data valid before and after execution.